H$_2$Open Journal

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H$_2$Open Journal Vol 8 No 5, 402 doi: 10.2166/h2oj.2025.002

Evaluation of water extraction indices for spatial mapping of surface water bodies using *Sentinel*-2: GIS and remote sensing approaches: the case of Ethiopia

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WG, 0009-0007-7606-6080

ABSTRACT

This study evaluates various water extraction indices to identify surface features using Sentinel-2 satellite images. Eight study sites were selected with three indices employed in this study, such as the water ratio index (WRI), the modified normalized difference water index (MNDWI), and the normalized difference water index (NDWI). The results of the accuracy assessment revealed that the producer accuracy (PA) for the NDWI ranged from 81.11 to 94.56%, that for the MNDWI ranged from 85.3 to 96.7%, and that for the WRI ranged from 96.98 to 97.98%. The user accuracies for the NDWI, MNDWI, and WRI results are 68.1–81.6%, 74.81–86.14%, and 80.33–88.78%, respectively. This study reveals that the WRI has greater PA (97.98%), user accuracy (88.78%), and kappa statistic (0.88) than the MNDWI and NDWI. The WRI is particularly suitable for extracting surface bodies surrounded by bare lands with minimal vegetation cover. Conversely, the NDWI is more applicable in areas with moderate vegetation and shadows, such as agricultural regions or wetlands. The MNDWI is suitable to avoid confusion water bodies with built-up areas and used to extract water bodies with various land cover. Therefore, establishing optimal thresholds is essential for policymakers and planners to effectively monitor water surface features and changes in water sources.

Key words: extraction indices, remote sensing, water body, water index

HIGHLIGHTS

- Addresses the extraction of surface water bodies under different environmental conditions.
- Utilizes water surface features separation indices with optimum threshold values.
- Reveals remote sensing and geographic information systems to identify and extract water bodies.

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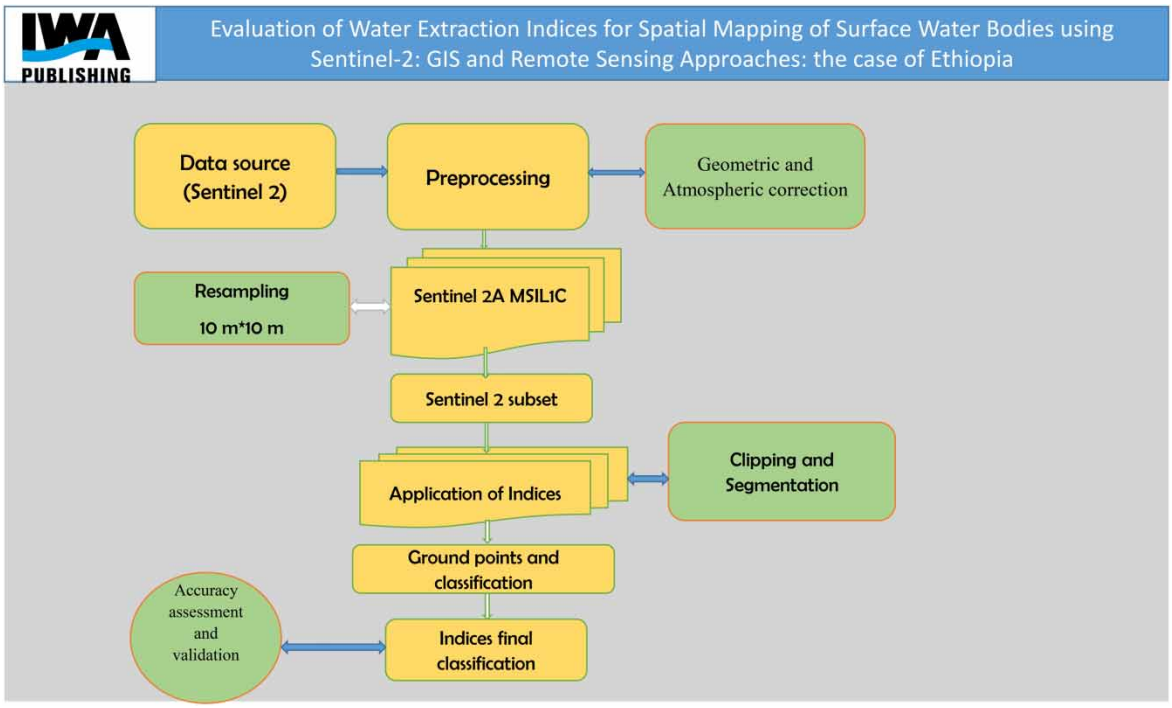
GRAPHICAL ABSTRACT



1. INTRODUCTION

Water plays a significant role in the sustainability of all living environments on Earth and can be found in a variety of forms, such as frozen water, surface water bodies, and groundwater. Other ecosystems are also highly dependent on this natural resource, which is crucial for their sustainability (Bernstein 2013). Remote sensing and geographic information systems (GIS) are approaches used to acquire and manipulate information about different water and non-water surface features using sensor technology (Zhong *et al.* 2018). The evaluation of various remote sensing and GIS-based approaches for surface water extraction has been crucial in enhancing surface water extraction and mapping (Laonamsai *et al.* 2023). Accurate remote sensing data and their application are highly important for the spatial mapping of surface water sources (Yan *et al.* 2023; Pham-Duc 2024). Therefore, the extraction and spatial mapping of surface water sources has become very important for the purposes of irrigated land mapping (Mohammed *et al.* 2023), change detection (Yoshe 2025), pumped hydro storages detection (Girma & Gebeyehu 2025), hydropower development (Athayde *et al.* 2019; Girma *et al.* 2025), hydrological modeling (Laonamsai *et al.* 2023), ecosystem modeling (Obahoundje *et al.* 2018), large-scale surface water change detection (Wieland & Martinis 2020), surface water source monitoring (Lekhak *et al.* 2023), flood mapping and assessment (Pena-Regueiro *et al.* 2020), and hydrological processes and water resources management (Jiang *et al.* 2021).

Currently, remote sensing approaches that utilize satellite images are widely used for the delineation of surface water bodies to extract data, such as water, plant, built-up area, and bare soil data, without requiring direct measurements (Ebadi 2022). Furthermore, these studies provide important insight into distinguishing between water and other land features, especially land covered by major lakes or rivers, where detecting changes may be challenging (Poortinga *et al.* 2019; Lekhak *et al.* 2023).

Numerous sensors and resolutions in satellite remote sensing are used to conduct accurate and timely environmental assessments and surface water monitoring, which are crucial for assessing hydrological processes and managing water resources (Jiang *et al.* 2021). Surface water can be distinguished from its surroundings by extracting water bodies via different indices from remotely sensed satellite images. The European Space Agency (ESA) provides Sentinel-2 imagery with remote sensing data to provide Earth observations that cover a wide geographic area and enable the detection of water bodies. Various approaches are widely employed for the extraction and spatial mapping of surface water bodies via satellite imagery. The threshold setting approach of the Otsu algorithm applies spectral indices such as the normalized difference water index (NDWI) (Bhangale *et al.* 2020), the

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modified normalized difference water index (MNDWI) (Yang & Chen 2017) and the water ratio index (WRI) (Khalid *et al.* 2021), and the automated water extraction index (AWEI) (Feyisa *et al.* 2014) has been used to investigate the performance or accuracy of various types of water surfaces. The method was used to improve water identification accuracy through the application of amendments to the NDWI, which was initially proposed (McFeeters 1996) to enhance the image of the NDWI. An MNDWI was provided by Xu (2006) by using the shortwave infrared (SWIR) band to replace the NIR band to improve the spectral difference between settlement land and water surfaces. Furthermore, when the NDWI and MNDWI are compared via satellite data obtained via remote sensing techniques, the MNDWI has an advantage over the NDWI in terms of lowering and even excluding settlement area noise, and it is more suitable for enhancing and delineating surface water (Xu 2006).

Moreover, studies have been conducted to introduce new approaches, such as that proposed by Li *et al.* (2021b), to automatically extract lakes via Sentinel-1 and Sentinel-2 multispectral images where a novel framework was developed comprising a robust water extraction network and a bimodal threshold segmentation approach. However, these methods have not been tested for complex weather conditions, built-up areas, or topographical features, but they have provided excellent results. The remote sensing stream turning Sentinel-2 multispectral imagery-based automatic approach was developed to extract complicated river networks for existing and new channels in different seasons, which revealed a more realistic separation of small rivers and streams, with some confusion between domestic water and other non-water surface features (Wang *et al.* 2021). The model is less time-consuming than the other methods. This model depicts segmentation and edge extraction events; however, further evaluation is needed, and it must be tested under diverse conditions.

The evaluation of GIS- and remote sensing-based water feature extraction indices is essential to alleviate the challenges related to extracting water features from site imagery. Although water feature extraction is simple because of the availability of widely used indices, the task becomes complex because of the variability of water bodies, such as sedimentation, built-up areas, and vegetation cover. The effectiveness of selected water feature extraction indices may be affected by the prevalence of vegetation cover. These methods may be more effective in areas with high vegetation cover, whereas other methods may be more effective in areas with low vegetation cover (Laonamsai *et al.* 2023). Three widely used approaches were selected for this study; however, notably, the performance and accuracy of these techniques may vary when they are implemented at different locations (Yang & Chen 2017; Laonamsai *et al.* 2023). Moreover, the selected water index-based approaches may depict various threshold values that may be higher or lower under specific conditions. Consequently, the use of single water indices, such as the NDWI, MNDWI, or WRI, may not be an optimal approach for the extraction of water sources. This may necessitate the comparison of multiple indices and the selection of the most suitable index for a given site and condition (Bhangale *et al.* 2020). Studies have used different approaches in Ethiopia, such as the NDWI, MNDWI, WRI, normalized difference moisture index, and AWEI. Studies were undertaken to assess water body detection approaches by considering various indices across Ethiopia's diverse geographical and topographical settings via high-resolution Sentinel-2 imagery and defined single threshold values for each feature and indices (Tesfaye & Breuer 2024); however, practically variable threshold values were expected with respect to diverse geographical and topographical settings. The Landsat 8 and Landsat 5 imagery was intended to detect changes around Adama in the extent of water bodies over time, but it has gaps in providing optimum thresholds and accuracy assessments (Sisay 2016; Sathianarayanan 2018). Change detection using multi-temporal imagery for the Rift Valley Lakes Basin results obtained from various indices such as NDWI and AWEI to detect how lake coverage changed within a defined period (Sisay 2016; Asfaw *et al.* 2020). A machine learning-based remote sensing approach was implemented to assess the spatiotemporal trends of surface water bodies at four study sites in Ethiopia. This approach provides spatiotemporal data for monitoring surface water bodies (Tesfaye & Breuer 2025). There is still variation in the findings regarding surface water extraction. It is difficult to establish well-defined threshold levels or limits, which demands further study.

Despite extensive prior studies in the country, critical research gaps remain regarding the detection and mapping of water bodies via remote sensing. This study addresses the limitations of determining and analyzing optimal threshold values for evaluating water extraction indices in areas across Ethiopia with diverse geographical and environmental conditions. While previous studies have examined different water indices using Sentinel-2 and Landsat imagery (Sisay 2016; Sathianarayanan 2018; Tesfaye & Breuer 2024), the determination of the optimal thresholds was either not addressed or addressed only in a limited way, which is crucial for the separation of water and non-water surface features. This research uniquely addresses this gap on optimal threshold determination and will examine how different indices (WRI, MNDWI, and NDWI) perform across varying sites,

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including natural lakes and artificial reservoirs. Given this motivation, the study addresses an important practical gap in optimum threshold values analysis to assess surface water resources by evaluating methodologies relevant to natural lakes, multipurpose reservoirs, and hydropower reservoirs monitoring. With Ethiopia's growing water resources development, there is a critical need for accurate and reliable methods to monitor water bodies by applying optimal thresholds that can be used in complex terrains and varied environmental conditions.

Accordingly, the objective of this study is to assess or evaluate approaches used to extract or delineate surface water bodies under different climates and topographic conditions of hydropower reservoirs and natural lakes in Ethiopia via different indices. The current study acquired satellite images from Sentinel-2, which have resolutions of 10 and 20 m and are available in a variety of electromagnetic spectrum bands, including blue, green, red, NIR, and Shortwave Infrared (SWIR). Extracting water features from satellite imagery requires quantifying the characteristics of water bodies such as lakes, rivers, reservoirs, and wetlands. It also requires the ability to identify and classify these features from remotely sensed data. This study has also addressed optimal threshold determination to evaluate various GIS-based water feature extraction indices to ensure the accuracy of the extracted data. The accuracy of the extracted data is very helpful for making informed decisions in these areas. Therefore, evaluating the performance of different indices is necessary to identify the most accurate and reliable methods for extracting water features. This research can help researchers determine which water surface feature information is needed for planning water resource development and sustainable water resource management in the area.

2. MATERIAL AND METHODS

2.1. Study area

In this study, hydropower reservoirs and natural lakes, such as Koka, Gibe I and III, Tekeze, GERD, Melkawakena, Genale Dawa III, Lake Tana, Lake Abaya-Chamo, Lake Hawassa, and the Central Lakes region, whose surroundings cover diverse surface water body types, were selected for evaluating the accuracy and extraction via Sentinel-2 imagery (Figure 1). The selected dam reservoirs and natural lakes situated in diverse geographical settings can be used to assess the efficiency and adaptability of the proposed method.

The study area is located in Ethiopia, a country characterized by diverse topography, including mountains, escarpments, rugged terrain, and low plains. In this study, surface waterbodies of hydropower reservoirs, manmade and artificial lakes, rivers, etc., are extracted. Geographically, Ethiopia is situated between latitudes of 3°37′ N and 15°35′ N and longitudes of 31°35′ E and 47°30′ E, the river basins of Ethiopia are classified into 12 of these 8 basins, which are occupied by rivers, one lake region and the remaining three dry basins (Berhanu *et al.* 2014). The country accounts for 99.3% of non-water land features, while 0.7% are covered by surface water features (Berehanu *et al.* 2014); furthermore, the country is characterized by different geological features, such as Precambrian basement, Paleozoic and Mesozoic sedimentary rocks, Tertiary sedimentary and volcanic rocks, and Quaternary sediments and volcanic rocks, with coverages of 18, 25, 40, and 17%, respectively (MoWR 2009).

2.2. Data acquisition and processing

The selection of study sites consisted of existing hydropower reservoirs and natural lake reservoirs, and the study sites were selected to use the results of this study for further analysis of the prospects of pumped storage hydropower for existing hydropower reservoirs and natural lakes. The evaluation of GIS and remote sensing approaches for spatial mapping of surface water bodies via Sentinel-2 and water indices could help researchers to understand and characterize hydropower reservoirs and lakes to promote sustainable, renewable energy development.

The main data used in this study were different bands of satellite images acquired from online sources of the ESA's Sentinel-2 satellite project. Furthermore, ground truth data were collected through consecutive site observations and support Google Earth.

The satellite images have 13 different spectral bands with variable resolutions, as depicted in Table 1. The near-infrared band (NIR) or Band 8 and the RGB color bands, or Bands 2 through 4, have a resolution of 10 m. These bands are used to calculate the WRI, MNDWI, and NDWI. There is a 20 m resolution in the vegetation red boundary bands. A spatial resolution of 20 or 60 m is available for the SWIR bands, namely, Band 10, Band 11, and Band 12.

The ESA launched the Sentinel-2 mission to provide high-resolution images of the surface of the Earth, which can be used for various purposes, including land cover classification, agricultural monitoring, and natural disaster

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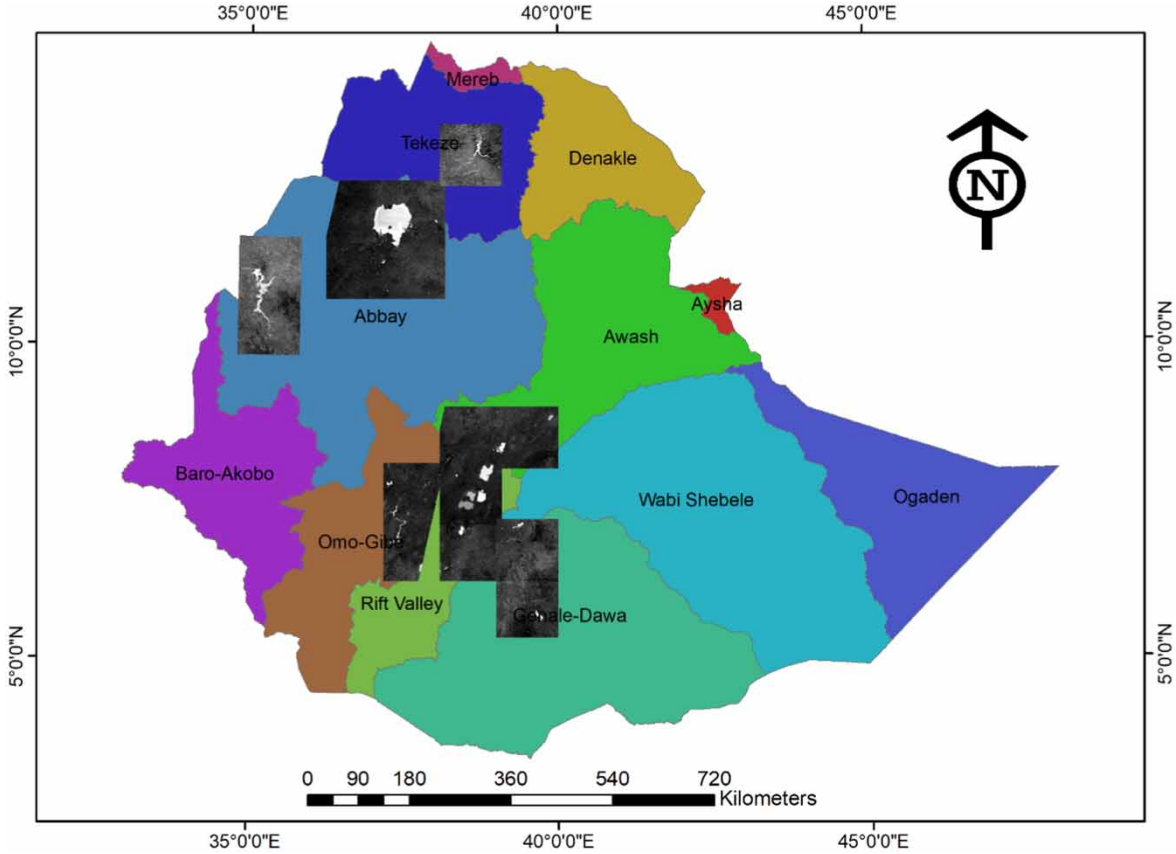

Figure 1 | Location map of the study area.

Table 1 | The list of bands with their central wavelengths and resolutions for the Sentinel-2 satellites

Bands (Resolution (m))	Band description
Band 1 (60)	Coastal aerosol
Band 2 (10)	Blue
Band 3 (10)	Green
Band 4 (10)	Red
Band 5 (20)	Vegetation Red Edge 1 (VRE1)
Band 6 (20)	Vegetation Red Edge 2 (VRE2)
Band 7 (20)	Vegetation Red Edge 3 (VRE3)
Band 8 (10)	Near-infrared (NIR)
Band 8A (20)	Vegetation Red Edge 4
Band 9 (60)	Water Vapor
Band 10 (60)	Shortwave Infrared 1 (SWIR 1)
Band 11 (20)	Shortwave Infrared 2 (SWIR 2)
Band 12 (20)	Shortwave Infrared 3 (SWIR 3)

Source: https://dataspace.copernicus.eu/analyse/apis/sentinel-hub (accessed January 1–15, 2023).

management. The raw data from the satellite has been carefully processed to ensure its accuracy, reliability, and usability for further analysis. The processing of Sentinel-2 satellite data is a preliminary step to use the datasets of downloaded images. It was conducted with geometric and atmospheric correction, creating image composites and mosaics, and resampling the data to a resolution of 10 m can ensure that the data are accurate, reliable, and usable for analysis. Geometric correction is a pre-processing step required to eliminate any biases or

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inaccuracies in the data caused by the satellite's sensor or its orbital path. This process adjusts the data to accurately reflect the true geographic coordinates of the Earth's surface. Atmospheric correction is required to remove the effects of the atmosphere on the data, such as light scattering or absorption, which can affect image accuracy.

The geometric and atmospheric inaccuracies were corrected, then subsequently, a composite and mosaic image was developed. This process involves merging several images captured by the satellite at various times and locations to form a continuous and comprehensive picture of the study area. Moreover, we resampled the data for defined resolution, specifically 10 m, utilizing specialized software like ArcGIS Pro 3.

The permanent water features are easily identifiable using the downloaded satellite imagery covering the study area. In addition, no cloud datasets were selected in any of the data subsets, which would have obscured the underlying surface features. The selection and acquisition of proper data highly depend on cloudy and rainy free weather conditions from 1 to 15 January to obtain cloudless Sentinel-2 images; however, other months, as they are affected by cloudy and rainy weather, make it difficult to obtain the required data. More cloud-free data are produced from imagery taken during dry seasons, which increases the accuracy of the data analysis (Sun *et al.* 2012; Jiang *et al.* 2014; Bishop-Taylor *et al.* 2019; Schmitt 2020).

ArcGIS Pro 3 was used to process Sentinel-2 satellite data, enabling the separation or extraction of water and non-water surface features. The use of ArcGIS Pro 3, a GIS software, enables the processing of Sentinel-2 satellite data and allows users to effectively manage and analyze the information using water extraction indices. The spatial analyst tools of ArcGIS Pro 3 provide a variety of functions and tools for processing and examining satellite data, including the ability to resample data to a desired resolution.

After having improved the atmospheric correction and resampling procedures, we faced some limitations of the approach. The cloud cover can significantly affect the accuracy of the information, and is noted as one of the challenges. The presence of clouds can hide the land surface and mislead the surface features, resulting in missing information or inaccurate reflectance values. Limitations in detection are caused by the color of water bodies and the limitations of remote sensing spectral information in complex geographic environments. Mountain shadows are often incorrectly identified as water bodies (Jiang *et al.* 2014; Li *et al.* 2022). Additionally, it is difficult to distinguish water from low-albedo pixels, such as shadows and asphalt (Feyisa *et al.* 2014). In some cases, a single pixel may contain more than one land cover type, resulting in the presence of mixed pixels. To mitigate this problem, we used spectral unmixing, which separates the reflectance values of each surface feature.

2.3. Multispectral indices

Currently, different water indices are utilized to identify and categorize water bodies via satellite images from Sentinel-2 satellites. These indices include the WRI, the MNDWI, the normalized difference vegetation index (NDVI), and the NDWI; furthermore, Otsu's thresholding approach was implemented on the index images of selected study sites with respect to each index.

The purpose of computing these indices is to improve the contrast between areas with different surface features. These indices aid in the isolation and highlighting of the distinctive spectral characteristics that separate water bodies from other land features.

In these indices, non-water classifications, such as soil, vegetation, and built-up regions, typically have negative values or values extremely close to zero, whereas water bodies typically have positive values (Xu 2006; Feyisa *et al.* 2014).

Currently, a variety of band indices are available for the purpose of water surface extraction and to differentiate non-water classes for multispectral data. These indices employ water and non-water classes or feature spectrum reflectances of different spectral bands.

The following data processing flowchart, which is depicted in Figure 2, shows the overall framework of the study.

Normalized difference water index (NDWI): The target item brightness level increases as the brightness value of the background objects is inhibited by employing a change in the reflectance values between the surfaces to be identified (Ji *et al.* 2009; Li *et al.* 2014). McFeeters developed the NDWI on the basis of this idea, utilizing Band 3 and Band 8 data taken from Sentinel-2 data, which may successfully suppress information about soil and vegetation and increase water availability (McFeeters 1996). The NDWI was developed to depict water bodies by using satellite images and estimates the changes between the green and NIR reflectance values. The NDWI

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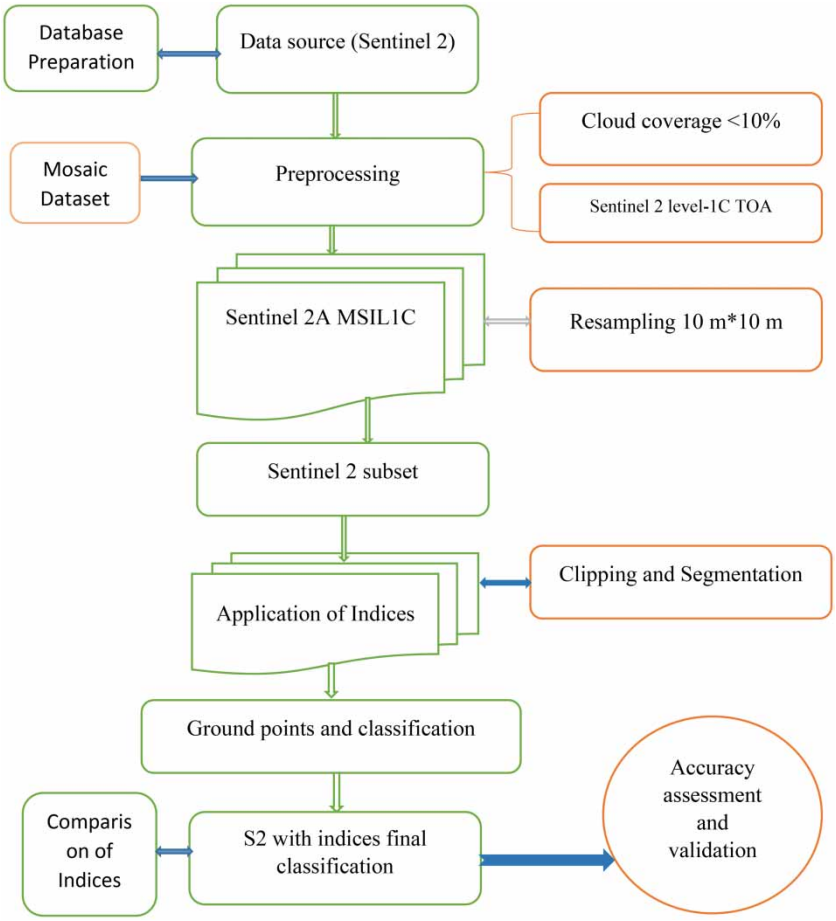

Figure 2 | Data processing flow chart.

was computed via Equation (1):

$$\mathrm{NDWI} = \frac{\mathrm{Band3_{Green}} - \mathrm{Band8_{NIR}}}{\mathrm{Band3_{Green}} + \mathrm{Band8_{NIR}}} \tag{1}$$

Green Band 3 depicts the reflectance of the green band (Band 3 of Sentinel-2) and represents the reflectance of the NIR band (Band 8).

Modified normalized difference water index (MNDWI): The modified NDWI (MNDWI) is known for enhancing surface water but is more efficient than the NDWI because of its ability to minimize and remove settlement area noise. Since the NIR band and the reflectance of built-up areas abruptly increase from the NIR to the SWIR band, Xu discovered that built-up areas might be confused with surface water extraction when the NDWI is used (Xu 2006). As a result, the MNDWI uses the SWIR band rather than the NIR band, and building noise can be successfully reduced to improve water bodies (Du *et al.* 2016). This index helps in the mapping of water bodies without the influence of vegetation. It was computed via Equation (2):

$$\mathrm{MNDWI} = \frac{\mathrm{Band3_{Green}} - \mathrm{Band11_{SWIR2}}}{\mathrm{Band3_{Green}} + \mathrm{Band11_{SWIR2}}} \tag{2}$$

The reflectance of Band 3 is represented by the formula Band 3_{Green}, and the reflectance of Band 11 is represented by Band 11_{SWIR2}. The NIR and green bands have a resolution of 10 m, whereas the SWIR2 band has a spatial resolution of 20 m.

Water ratio index (WRI): The WRI is an important parameter used to extract and map surface water information. This index was developed to compare surface water types and their health for water bodies with

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different reflectance values. Water has greater spectral reflectance in the red (Band 4) and green (Band 3) bands than in the NIR (Band 8) and medium-infrared (Band 5) bands; therefore, WRIs result in values greater than 1.00 (Shen & Li 2010).

$$\mathrm{WRI} = \frac{\mathrm{Band3_{Green}} + \mathrm{Band4_{Red}}}{\mathrm{Band8_{NIR}} + \mathrm{Band11_{SWIR2}}} \tag{3}$$

The above-mentioned water indices may simplify the work of satellite image processing to classify water bodies and non-water features, allowing accurate mapping and monitoring of overall water resources for further study and decision-making in different fields of study.

2.4. Accuracy assessment

To validate and assess accuracy, random sample points and collected ground control points (randomly collected for non-water surface features and surface water features representative of ground points) were used for each study area. Furthermore, the results are checked to ensure that each random sample point is accurate and that ground control points are created and overlaid. Topo-maps and Google Earth were utilized as the reference data. For the purpose of validation, a total of 200 sample sites were chosen at random from recent water bodies in the field. Accuracy verification is carried out via visual interpretation and quantitative accuracy approaches. The accuracy of each index was assessed by applying the following mechanisms: producer accuracy (PA), user accuracy (UA), overall accuracy (OA), and the kappa coefficient (KPC).

Producer accuracy: This metric estimates how many true positive water cells can be expected in comparison with the number of water cells actually present.

User accuracy: This metric measures the probability of correctly labeling a pixel as water, considering all the pixels classified as water (Equation (5)).

Overall accuracy: This matches the universal accord imposed on every specific determination (Equation (6)).

Kappa coefficient: This coefficient examines the reciprocal adjustment in excess or deficit as it appears from actual use and calculated design (Equation (7)).

$$\mathrm{PA} = \text{True Positives (TP)}/(\text{True Positives (TP)} + \text{False Negatives (FN)}) \tag{4}$$

$$\mathrm{UA} = \text{True Positives (TP)}/(\text{True Positives (TP)} + \text{False Positives (FP)}) \tag{5}$$

$$\mathrm{OA} = \text{Total Correct Predictions}/\text{Total Predictions} = \sum \mathrm{TP}/\sum(\mathrm{TP} + \mathrm{FP} + \mathrm{FN}) \tag{6}$$

$$\mathrm{KPC} = \frac{T \times (\mathrm{TP} + \mathrm{TN}) - (\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN}) \times (\mathrm{FP} + \mathrm{TN})}{T \times T - (\mathrm{TP} + \mathrm{FP}) \times (\mathrm{TP} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TN}) \times (\mathrm{FP} + \mathrm{TN})} \tag{7}$$

where T is the number of pixels among the total values, TP denotes true positives, TN denotes true negatives, FN denotes false negatives, and FP denotes false positives.

3. RESULTS

3.1. Optimum thresholds for separation of surface water features

Optimum threshold values were determined for each water and non-water surface feature and index in the following locations: the GERD and Tana in the Abay Basin; Gibe I and III in the Omo Gibe Basin; Koka in the Awash Basin; Tekeze in the Tekeze Basin; Melkawakena in the Wabishebele Basin; Genale Dawa III in the Genale Dawa Basin; the Central Lakes; Lake Hawassa; and Abaya-Chamo in the Rift Valley Lakes.

Figure 3 depicts the results of the analysis of the separation by applying threshold values of surface water features through the NDWI, MNDWI, and WRI. Each water index was evaluated for its suitability for application to each selected study area. The three water indices identified were the NDWI, the MNDWI, and the WRI. The threshold method of Otsu's approach is designed to detect boundary limits (Ji *et al.* 2009; Sekertekin 2019; Yang *et al.* 2020) between two different classes, extract surface water bodies and non-water surfaces for image pixels that have similar or related values and is applied to images of indices with specified study areas, such as dam reservoirs, natural lakes, and surrounding streams (Sekertekin 2021).

The separability of different surface features revealed that variable threshold values with low values represent non-water features, and higher values represent surface water because the unique spectral properties of water

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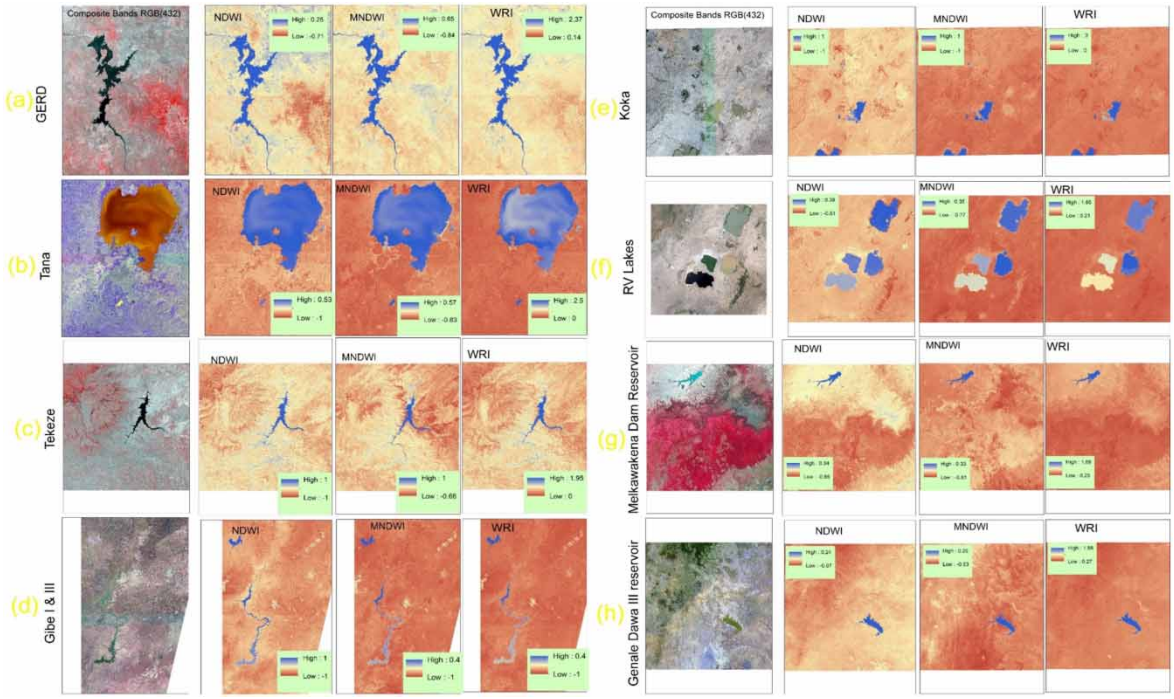

Figure 3 | Water body extraction indices from left to right composite images (R: Band 4; G: Band 3; B: Band 2), NDWI, MNDWI, and WRI for the (a) GERD, (b) Tana Reservoir, (c) Tekeze Reservoir, (d) Gibe I and II Reservoirs, (e) Koka Reservoir, (f) Central Rift Valley Lakes, (g) Melkawakena Reservoir, and (h) Genale Dawa III Reservoir.

absorb most of the solar radiation in the NIR region, resulting in low reflectance values. In comparison, however, there are more strong reflections for the green waveband caused by light scattered by water vapor.

As shown in Table 2, the threshold values range from 0 to 0.27, 0.02 to 0.32, and 0.90 to 1.4, indicating surface features for the NDWI, MNDWI, and WRI, respectively. The MNDWI values from −0.47 to −0.26, −0.35 to 0.20, and 0.26 to 0.65 represent the vegetation surface features of the NDWI, MNDWI, and WRI, respectively.

A comparison of the threshold values for built-up surface features is shown. The MNDWIs, WRIs, and NDWIs range from −0.3 to −0.12, −0.25 to −0.12, and 0.56 to 0.76, respectively.

Table 2 | Threshold values for water bodies and non-water features

	Threshold values (Water)			Threshold values (Vegetation)			Threshold values (Built-up)		
Selected study areas	**NDWI**	**MNDWI**	**WRI**	**NDWI**	**MNDWI**	**WRI**	**NDWI**	**MNDWI**	**WRI**
GERD	0.03	0.05	0.95	−0.35	−0.20	0.34	−0.300	−0.150	0.600
Tana Lake	0.27	0.32	1.40	−0.47	−0.30	0.40	−0.210	−0.250	0.560
Gibe I and III	0.05	0.07	1.10	−0.34	−0.29	0.40	−0.200	−0.140	0.580
Awash (Koka)	0.02	0.08	1.15	−0.37	−0.31	0.45	−0.190	−0.170	0.650
Tekeze	0.00	0.02	0.90	−0.26	−0.32	0.65	−0.190	−0.170	0.650
Melkawakena	0.01	0.04	1.10	−0.45	−0.30	0.44	−0.120	−0.130	0.700
GD-III	0.11	0.15	1.15	−0.37	−0.27	0.41	−0.150	−0.190	0.700
Central Lakes	0.01	0.03	1.04	−0.26	−0.32	0.57	−0.180	−0.140	0.760
Hawassa Lake	0.01	0.05	1.12	−0.46	−0.35	0.40	−0.170	−0.160	0.750
Abaya and Chamo Lakes	0.03	0.03	0.90	−0.46	−0.33	0.26	−0.160	−0.120	0.570
Average	0.05	0.08	1.08	−0.38	−0.30	0.44	−0.18	−0.17	0.67
Var.	0.01	0.01	0.02	0.00	0.00	0.01	0.00	0.00	0.01
Std.	0.07	0.08	0.13	0.07	0.04	0.10	0.05	0.04	0.09

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The NDWIs, MNDWIs, and WRIs represent threshold value increments of built-up, vegetation, and water surface features, respectively. All these threshold values indicate that water surface features have higher values than do built-up areas and vegetation.

The bands considered in the extraction of surface water features by the threshold values of each index with respect to each surface feature tend to overlap when the spectral responses at several wavelengths are quite comparable, such as cultivated land features, shrub land, vegetation cover, and built-up areas, where features separated might depict similar values at various wavelengths. Visible separability of features was observed in highly built-up areas and water surfaces (with good quality). We also identified the thresholds of different land features, such as built-up areas, vegetation, and water surface features. On the basis of the identified threshold values, the possibility of class identification and separation of surface features can be attained. However, thresholds of each feature, in which the features can be separated, might overlap significantly under some conditions.

3.2. Water mapping results

The images for the selected study areas, including Koka, Gibe I and III, Tekeze, GERD, Melkawakena, Genale Dawa III, Lake Tana, Lake Abaya-Chamo, Lake Hawassa, and Central Lakes Region reservoirs, encompass a range of surface water bodies. The aforementioned surface water extraction indices were processed via the NDWI, MNDWI, and WRI approaches, which clearly indicate where large bodies (natural lakes and manmade reservoirs) and rivers exist, as depicted in Figure 3. The NDWI-, MNDWI-, and WRI-based approaches can extract most of the water surface areas for the selected study area.

3.2.1. Normalized difference water index

The NDWI is used for assessing, detecting, and mapping surface water sources of selected study areas of Koka, Gibe I and III, GERD, Lake Tana, Melkawakena, Genale Dawa III, Lake Abaya-Chamo, Lake Hawassa, the Central Lakes region, and Tekeze Reservoirs with their surrounding non-water surface features. The results indicate that the NDWI values of water surface features range from 0.01 to 0.27; more enhanced water surface information results in higher NDWI values of 0.01 with respect to other surface features (Figure 4).

The NDWI values of the built-up land (minimum −0.33 and maximum −0.22) and vegetation (minimum −0.47 and maximum −0.26) areas are shown in Figure 4. The results show that water surface features have much higher NDWI values than vegetation features do; these values can be easily extracted from water bodies that have the

	GERD	Lake Tana	Gibe I & III	Awash (Koka)	Tekeze	Melkawakena (Wabish)	GD_III	Central Lakes	Hawasa Central Lakes	Abaya Lakes
Min	-0.7	-0.66	-1	-1	-1.00	-0.66	-0.67	-0.62	-0.69	-0.66
Max	0.25	1	1	1	1.00	0.34	0.25	0.39	0.36	0.53
Mean	-0.24	-0.31	-0.31	-0.25	-0.26	-0.36	-0.32	-0.22	-0.37	-0.25

	GERD	Lake Tana	Gibe I & III	Awash (Koka)	Tekeze	Melkawakena (Wabish)	GD_III	Central Lakes	Hawasa Central Lakes	Abaya Lakes
Water	0.03	0.27	0.05	0.02	0.00	0.01	0.11	0.01	0.01	0.03
Vegetation	-0.35	-0.47	-0.34	-0.37	-0.26	-0.45	-0.37	-0.26	-0.46	-0.46
Built ups	-0.3	-0.21	-0.2	-0.19	-0.19	-0.12	-0.15	-0.18	-0.17	-0.16
Bare land	-0.1	-0.21	-0.16	-0.11	-0.16	-0.1	-0.05	-0.22	-0.2	-0.3

Figure 4 | (a) Mean, minimum, and maximum values of the NDWI and (b) NDWI values of water and non-water surface features.

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highest optimum threshold values of the NDWI (minimum 0.01 and maximum 0.27), which makes water bodies easy to distinguish from vegetation and other features. Most NDWI values of water bodies are greater than 0.01, indicating the unique spectral properties of water. Water absorbs most of the solar radiation in the NIR or Band 8, as depicted in Table 2, resulting in low reflectance values. In comparison, however, there are stronger reflections for the green waveband caused by light scattered by water vapor. Compared with air, greens have greater reflection because they scatter light from water particles. Our study depicted that NDWI values ranged from 0.01 to 0.27, with an average value of 0.05, in order to extract water resources from Sentinel-2 imagery.

3.2.2. Modified normalized difference water index

The MNDWI is applied in selected study areas, such as Koka, Gibe I and III, GERD, Lake Tana, Melkawakena, Genale Dawa III, Lake Abaya-Chamo, Lake Hawassa, the Central Lakes region, and Tekeze Reservoirs, with their surroundings for the purpose of assessing, detecting, and mapping surface water sources. As a result, Figure 5 shows that the values of the MNDWI for water bodies are typically greater than 0.02 (the minimum for the Tekeze Reservoir is 0.02, and the maximum for Lake Abaya-Chamo is 0.37), which is a threshold value used to identify and separate water bodies from non-water areas.

The MNDWI values are shown in Figure 5 for the vegetation (minimum −0.35 and maximum −0.2), built-up (minimum −0.25 and maximum −0.12), and water values of the NDWI (minimum 0.02 and maximum 0.32). The results show that water surface features have the highest optimum values for the MNDWI and can be easily extracted from other features and vegetation. Vegetation has significantly lower MNDWI values than water bodies do, making vegetation easy to distinguish from water bodies.

3.2.3. Water ratio index

The WRI was proposed for the purpose of assessing, detecting, and mapping surface water sources in selected study areas, such as Koka, Gibe I and III, GERD, Lake Tana, Melkawakena, Genale Dawa III, Lake Abaya-Chamo, Lake Hawassa, the Central Lakes region, and Tekeze Reservoirs, with their surrounding non-water surface features. As shown in Figure 6, most surface water features have WRI values greater than 0.9, which indicates that as the quality of water increases, the WRI values also increase, indicating the generally good health of our aquatic ecosystems (Gautam *et al.* 2015; Khalid *et al.* 2021; Laonamsai *et al.* 2023).

WRI values of the vegetation (minimum 0.26 and maximum 0.65), built-up (minimum 0.56 and maximum 0.85), and WRI values of the water (minimum 0.9 and maximum 1.4) areas. Furthermore, vegetation has

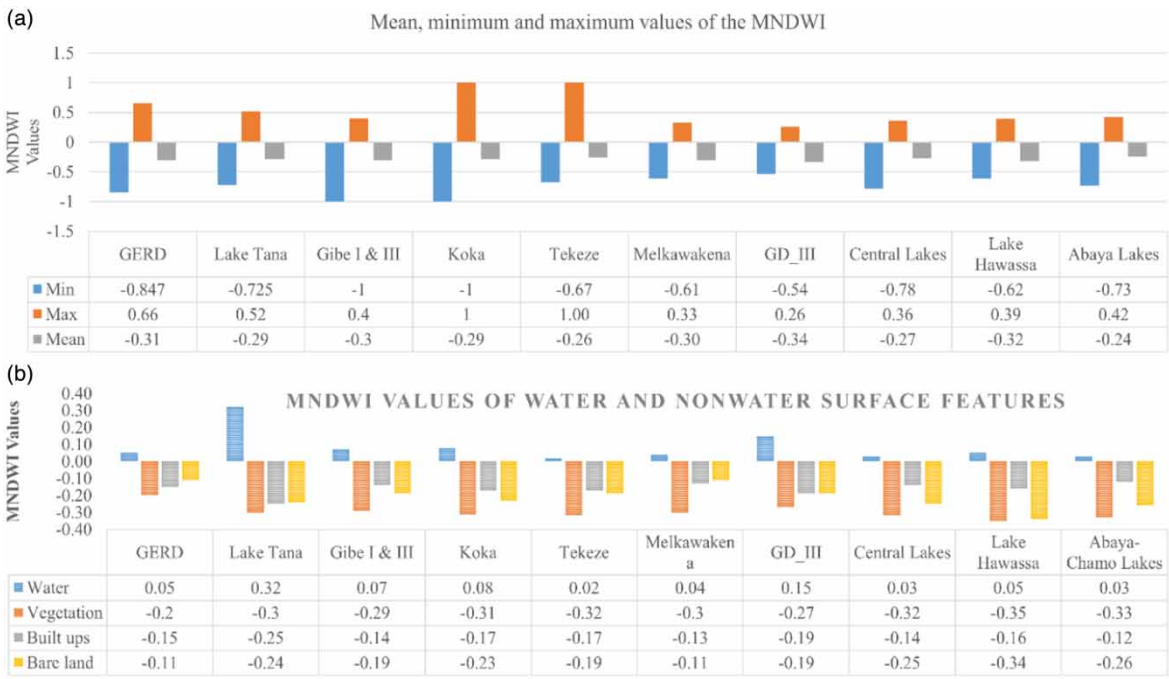

Figure 5 | (a) Mean, minimum, and maximum values of the MNDWI and (b) MNDWI values of water and non-water surface features.

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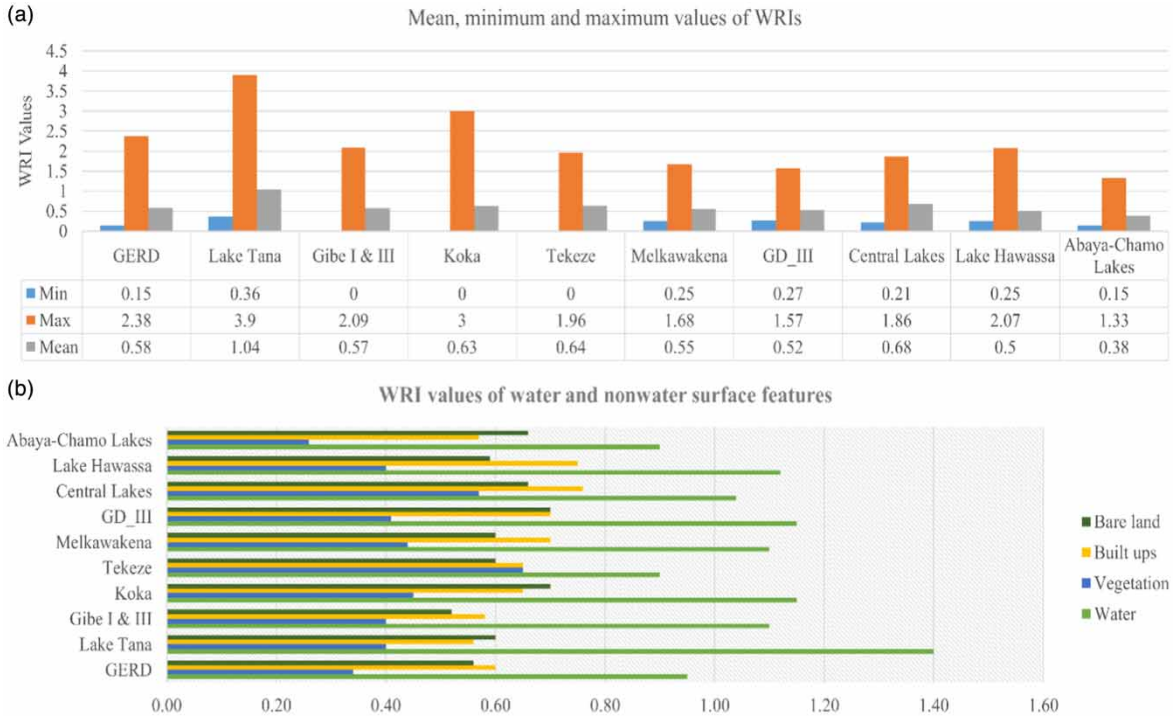

Figure 6 | (a) Mean, minimum, and maximum values of the WRI and (b) WRI values of water and non-water surface features.

much smaller WRIs than surface features do, which may lead to the easy identification of vegetation cover or features easily from water bodies; conversely, water surface features have the highest optimum threshold values of WRIs, which makes water bodies easy to distinguish from vegetation and other features.

3.2.4. Comparisons of indices

The results from indices depict values for all selected study sites for the following lakes and reservoir sites: Tekeze, Melkawakena, Central Lakes, Hawassa Central Lakes, Koka, Abaya Lakes, GERD, Gibe I and III, Genale Dawa III, and Lake Tana. The water extraction indices of selected lakes and reservoirs in Ethiopia were analyzed, and the range of values was depicted to extract various surface features.

The surface water feature values of the NDWIs, MNDWIs, and WRIs vary from 0 to 0.27, 0.02 to 0.32, and 0.90 to 1.4, respectively. Compared with those of the MNDWI and WRI, the lowest value range is found in the NDWI.

As shown in Table 2, the threshold values for the vegetation features also vary, ranging from −0.47 to 0.26, −0.35 to 0.20, and 0.26 to 0.65 for the NDWI, MNDWI, and WRI, respectively. A threshold value comparison of built-up surface features is presented. The MNDWI, WRI, and NDWI range from −0.3 to −0.12, −0.25 to −0.12, and 0.56 to 0.76, respectively. The NDWIs, MNDWIs, and WRIs represent threshold value increments of built-up, vegetation, and water surface features, respectively. The threshold values indicate that water surface features have higher values than do built-up areas and vegetation.

Furthermore, as shown in Figure 7, the lowest value of the NDWI is 0.00, the value obtained from Tekeze, whereas the highest value is 0.27, as shown in Lake Tana. Melkawakena, Lake Hawassa, Central Lakes, and Koka have relatively high NDWI values, ranging from 0.01 to 0.02. Abaya Lakes, GERD, and Gibe I and III have moderate NDWI values, ranging from 0.02 to 0.05, whereas GD-III has a relatively high value of 0.11.

Compared with the NDWI, the MNDWI is a better minimum threshold than the NDWI for enhancing surface water body information. The lowest value of the MNDWI is 0.02, which is lower than the NDWI values observed in Tekeze, and the highest value of 0.32 is obtained from Lake Tana. Melkawakena, Lake Hawassa, and Koka have relatively high MNDWI values, ranging from 0.04 to 0.08. The moderate MNDWI values depicted in Abaya-Chamo Lakes, GERD, and Gibe I and III range from 0.03 to 0.07, whereas Genale Dawa III has a comparatively high threshold value of 0.15. Compared with the NDWI, the MNDWI is better at enhancing surface water features. This is because the reflectance value interval between Band 3 and Band 11 or the SWIR band

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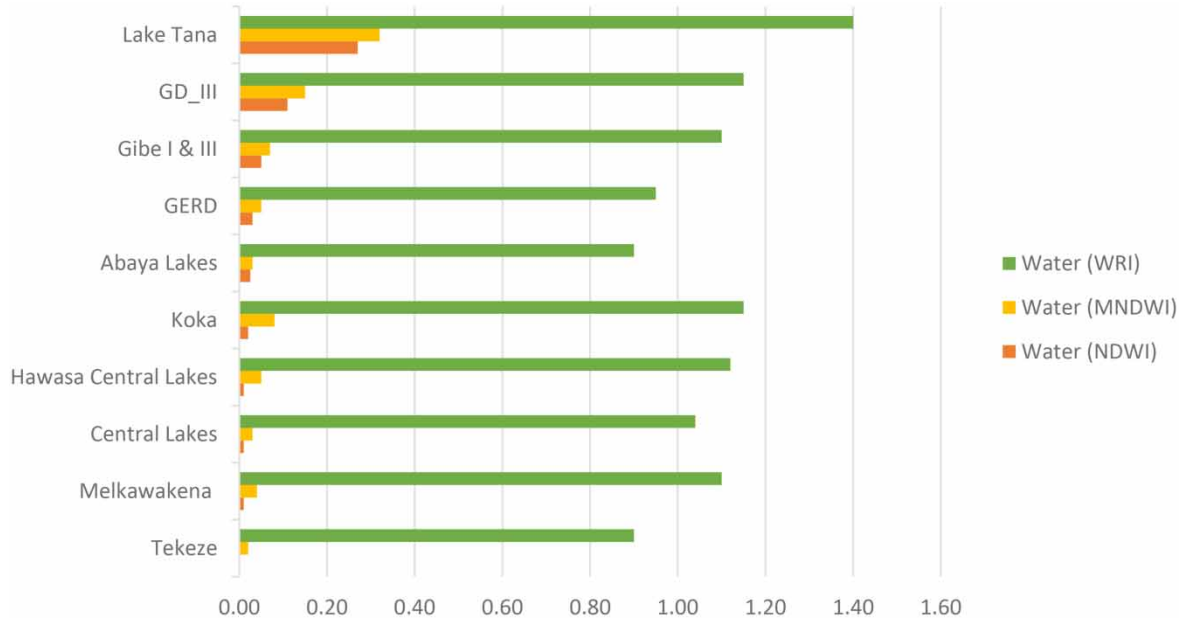

Figure 7 | NDWI, MNDWI, and WRI values for water surface features with respect to the selected study area.

is relatively larger than that between the green band (Band 3) and the NIR band (Band 8), which maximizes the optimal threshold of the MNDWI.

Water bodies have MNDWI values greater than zero and positive values greater than those of built-up regions, which have negative values that approach zero because the reflectance is greater for SWIR light than for green light. This is because they normally absorb more SWIR light than NIR light (Sun *et al.* 2012). The results show that water surfaces typically indicate low reflectance in the SWIR band and that high reflectance in the green band (Band 3) produces higher MNDWI values.

The WRI, on the other hand, has a relatively higher value than the NDWI and MNDWI. Lake Tana has the highest value, 1.4, whereas the lowest value, 0.9, is obtained from the Tekeze Reservoir. The WRI values for Koka, Lake Hawassa, and Melkawakena ranged from 1.10 to 1.15. Additionally, values ranging from 0.95 to 1.10 were observed for Gibe I and III, Genale Dawa III, and GERD. In particular, Lake Tana has a reasonably high WRI value of 1.4. This may be due to various factors, such as the relatively better quality of water and the lack of sedimentation, indicating the generally good health of our aquatic ecosystems.

3.3. Accuracy assessment

The evaluations were conducted on the basis of the optimum thresholds of Otsu's approach to evaluate the three water indices for the selected study sites, as depicted in Table 2. A summary of the different water types was provided with respect to each index.

Table 3 lists the quantitative accuracy approaches that were used to better examine the water extraction indices for the selected study sites and compares all three approaches to correctly classify various water and non-water bodies. Three methods, the NDWI, MNDWI, and WRI, were applied using Sentinel-2 data to determine the classification accuracy. With respect to water bodies, the NDWI depicts the accuracy of producers, with a minimum accuracy of 81.11% (Abaya-Chamo Lakes) and a maximum accuracy of 94.56% (Melkawakena); the MNDWI depicts a minimum accuracy of 85.3% (GD-III) and a maximum accuracy of 96.7% (Koka); and the WRI depicts a minimum accuracy of 96.98% (GERD) and a maximum accuracy of 97.98% (Abaya-Chamo Lakes). The kappa coefficients for the NDWI, MNDWI, and WRI are within the intervals of 0.63–0.88, 0.73–0.85, and 0.78–0.88, respectively.

The OA for each of the selected study areas ranged from 82.82 to 88.6% (NDWI), 88.42 to 93.66% (MNDWI), and 92.45 to 94.43% (WRI). The user accuracies for the NDWI, MNDWI, and WRI results are 68.1–81.6%, 74.81–86.14%, and 80.33–88.78%, respectively. Among the above methods, WRI first and MNDWI were the second most accurate, whereas the least accurate classification method was the NDWI method.

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Table 3 | Accuracy assessment for the selected study area

S.No.	Study area	Indices	Producers accuracy		Users accuracy		Overall precision accuracy	Kappa coefficient
			Water (%)	Non-water	Water	Non-water		
1	GERD	NDWI	86.71	84.23	81.6	86.43	88	0.63
		MNDWI	95.7	91	86.14	87.34	93.66	0.73
		WRI	96.98	91.37	88.78	92.47	94.43	0.78
2	Tana	NDWI	89.01	81.89	72.46	87.02	87.41	0.68
		MNDWI	95.00	89.42	78.72	88.76	94.61	0.78
		WRI	97.60	91.17	89.48	93.27	95.13	0.80
3	Gibe I and II	NDWI	90.57	81.79	68.36	87.12	88.6	0.69
		MNDWI	96.6	89.77	75.37	89.11	90.2	0.8
		WRI	97.46	88.98	80.85	92.95	93.95	0.84
4	Awash (Koka)	NDWI	91.23	81.94	68.21	89.44	87.16	0.72
		MNDWI	96.7	89.87	75.27	92.56	91.02	0.81
		WRI	97.58	89.1	80.73	95.07	93.83	0.87
5	Tekeze	NDWI	89.52	82.05	68.1	88.55	84.5	0.74
		MNDWI	95.3	89.96	75.18	93.68	92.7	0.82
		WRI	97.68	89.2	80.63	94.17	93.73	0.86
6	GD-III	NDWI	94.56	84.29	71.86	84.79	84.88	0.81
		MNDWI	85.3	82.19	74.96	95.52	89.6	0.76
		WRI	97.76	89.28	80.55	96.25	93.65	0.85
7	Melkawakena	NDWI	92.23	82.22	74.93	85.72	86.88	0.83
		MNDWI	94.56	89.99	75.15	95.33	88.42	0.82
		WRI	97.82	89.34	80.49	96.31	92.66	0.85
8	Central Rift Valley Lakes	NDWI	86.21	82.28	71.87	86.78	82.82	0.72
		MNDWI	96.06	90.23	74.91	94.57	90.65	0.84
		WRI	97.88	89.4	80.43	96.37	92.45	0.87
9	Hawasa Lakes	NDWI	87.56	82.34	70.81	87.84	83.68	0.72
		MNDWI	96.12	90.28	74.86	95	91.75	0.85
		WRI	97.93	89.45	80.38	96.82	93.48	0.88
10	Abaya Lakes	NDWI	81.11	82.39	71.26	84.39	86.71	0.73
		MNDWI	96.12	90.33	74.81	94.05	91.66	0.85
		WRI	97.98	89.5	80.33	96.47	94.42	0.86
Min		NDWI	81.11	81.79	68.10	84.39	82.82	0.63
		MNDWI	85.30	82.19	74.81	87.34	88.42	0.73
		WRI	96.98	88.98	80.33	92.47	92.45	0.78
Max		NDWI	94.56	84.29	81.60	89.44	88.60	0.83
		MNDWI	96.70	91.00	86.14	95.52	94.61	0.85
		WRI	97.98	91.37	89.48	96.82	94.43	0.88
Average		NDWI	88.87	82.54	71.95	86.81	86.06	0.73
		MNDWI	94.75	89.30	76.54	92.59	91.43	0.81
		WRI	97.67	89.68	82.27	95.01	93.77	0.85

Furthermore, Table 3 shows the OA and the KPC, revealing that the performance of the water indices (NDWI) resulted in 85.91% OA and 0.73% KPC, and the MNDWI approach resulted in 91.07% OA and 0.81% KPC. Compared with the NDWI and MNDWI, the WRI achieves improved performance in terms of 93.62% overall precision accuracy and 0.85% KPC. On the other hand, the OAs of the MNDWI and WRI in the GERD and Central Rift Valley Lakes were 93.66 and 92.45%, respectively. Additionally, the OAs of the MNDWI and WRI in the GERD and GD-III groups were similar at 93.66%. This finding indicates that the combination of proposed water indices might yield better results in separating surface water features.

4. DISCUSSION

This study aimed to evaluate different water surface feature extraction approaches for selected study sites in Ethiopia that feature variable topographic and weather conditions. Consequently, the findings of this study

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suggest the use of spatial mapping of surface water bodies to extract hydropower reservoir sites and natural lakes through the application of satellite images with surface water extraction indices. The easily accessible satellite data and surface water extraction approaches make GIS-based water surface extraction very popular. It addresses the evaluation of water surface extraction approaches and analysis of the obtained results in terms of different water extraction indices. Water extraction indices, particularly the NDWI, MNDWI, and WRI, are used in Otsu's approach to characterize optimal threshold values with respect to indices and with respect to different natural lakes and dam reservoirs. This study preceded the evaluation of water indices by defining optimal thresholds based on the behavior of different geographical and topographical settings. These indices are more effective in detecting surface water in diverse geographical and topographical settings. The study addresses the characterization of optimum threshold values, a topic that has not been sufficiently addressed in previous studies conducted in the country (Sisay 2016; Sathianarayanan 2018; Tesfaye & Breuer 2024).

This study found that applying water indices, a widely used approach to identifying and extracting surface water features, and the results of the assessment showed that the WRIs were more accurate than MNDWI and NDWI in detecting surface water (Mahdianpari *et al.* 2019; Li *et al.* 2021a). Furthermore, the study addressed characterization of optimal thresholds in terms of each site and indices, while other studies observed discrepancies with the thresholds that affect surface water separation in achieving well-defined approaches.

The accuracy of the findings varied across the selected study sites when all the study sites were considered. The lowest OA values depicted were NDWI (82.82%), MNDWI (88.42%), and WRI (92.45%) for the study areas of Central Rift Valley Lakes, Melkawakena, and Central Rift Valley Lakes, respectively. The OA values obtained in this study for water in the Rift Valley Lakes are acceptable for NDWI and MNDWI. There is a slight difference from previous studies (Asfaw *et al.* 2020). The maximum values of OA were the NDWI (88.60%), MNDWI (93.66%), and WRI (94.43%) for the selected study areas of Gibe I and III and GERD, respectively. In this study, this is clearly observed in Table 3, in which the MNDWI and WRI accuracy assessment values were higher than the NDWI values with respect to the selected study areas (Tesfaye & Breuer 2024). The results accuracy assessment of this study is usual and can be considered acceptable for evaluating the results of surface water detection using water indices can be reinforced with the outcomes of studies and are in agreement with the findings of studies (Yang *et al.* 2017; Khalid *et al.* 2021; Li *et al.* 2021a; Lekhak *et al.* 2023). Regarding the KPC from the water extraction indices of the NDWI, MNDWI, and WRI, Melkawakena and Lake Hawassa had the highest KPC compared with the other study sites (Asfaw *et al.* 2020).

Furthermore, when interpreting the performance or accuracy assessment for three water indices for the selected study sites, as shown in Table 3, for example, Melkawakena has the maximum PA compared with the minimum accuracy of 81.11% (Abaya-Chamo Lakes). This is because the surface features sediments and water without sediments (Tavares *et al.* 2019).

The maximum PA values for the three indices are as follows: NDWI is 94.56% at GD-III, MNDWI is 96.7% at Koka, and WRI is 97.98% at Abaya Lakes. However, the findings varied across the selected study sites. Previously studies were undertaken to assess water body detection approaches by considering various indices across Ethiopia's Sentinel-2 imagery and using accuracy assessment approaches mentioned in Section 2.4, except KPC by defining single threshold values for each feature and indices. The outcomes of the aforementioned PA differ slightly from those of previous studies, as indicated by Tesfaye & Breuer (2024): NDWI (85%), MNDWI (92%), and WRI (96%). The Landsat imagery was used to detect changes in water body coverage over time within a defined period. However, no accuracy assessment or performance evaluation of the indices was included (Sisay 2016; Sathianarayanan 2018).

In terms of the extraction of surface water features, the MNDWI and WRI were preferable to the NDWI, which demands the application of a combination of indices for the extraction of surface water sources. Our findings indicate that for areas covered with water surfaces, NDWI values greater than 0.01 can be used to identify areas of high water accumulation and for some lakes/reservoirs greater than 0.27, enabling the outputs to be utilized for further studies, and decision-making can be supplemented with the findings of studies (Yang & Chen 2017; Mahdianpari *et al.* 2019; Khalid *et al.* 2021). The NDWI can be applied without difficulty to the satellite images to extract water features from the terrestrial surface. In the context of Ethiopia, it has been found NDWI ranges from 0.01 to 0.27 with average value of (0.05), variable values obtained due to diverse environmental conditions, and solve difficulties arising in assigning single values for large-scale studies (Tesfaye & Breuer 2024).

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The results show that the NDWI is less effective at enhancing surface water body information when the MNDWI and WRI are compared; however it better represents the areas' shadow of built-up areas and bare land (Zhang *et al.* 2011). The MNDWI also has advantages over the NDWI for extracting surface water features, and it performs better when it is utilized for surface water (Bie *et al.* 2020). In our study, this is clearly shown in Table 3, in which the NDWI values were greater than 0.01 for the selected study sites of Tekeze, the central lakes, and Lake Hawassa, which can be seen in studies conducted in different parts of the world (Li *et al.* 2022; Laonamsai *et al.* 2023). Moreover, the average normalized difference water indices exceed the value of 0.01 and varied from 0.01 to 0.27, showing how useful it has proven itself as a method of checking the water content. The significant values for these indices above 0.01 are evidence of their ability to detect reservoirs accurately. This occurred because the reflectance value interval of the green band (Band 3) and NIR band (Band 8) is relatively less than that between Band 3 and Band 11 or the SWIR band, which reduces the optimum threshold value of the MNDWI, which may exceed the NDWI.

There are still some matters with respect to the selected study areas, revealing some confusion between water surface features and built-up areas, such as Koka, Tekeze, Abaya, Abijata, water bodies mixed with built-up areas, and NDWI values have been used to identify areas with high water contents. A higher NDWI value reveals greater water accumulation, but some confusion may occur between surface features (McFeeters 1996; Lu *et al.* 2011; Zhang *et al.* 2011; Jiang *et al.* 2014; Wang *et al.* 2018; Yang *et al.* 2018). However, Rokni *et al.* (2014) investigated modeling of temporal lake area change, the NDWI performed better than other indices and the findings disagree with our study while using the multi-temporal Landsat 5-TM, 7-ETM+, and 8-OLI images.

The results show that the MNDWI is better than the NDWI for enhancing surface water body information compared with the NDWI. This occurred because the reflectance value interval between Band 3 and Band 11 of the SWIR band is relatively greater than that between the green band (Band 3) and the NIR band (Band 8), which maximizes the optimum threshold value of the MNDWI; however, the NDWI performs better for water surface feature extraction in built-up areas (Jiang *et al.* 2014).

Water bodies have MNDWI values greater than zero and positive values greater than those of built-up regions, which have negative values that approach zero because the reflectance is greater for SWIR light than for green light. This is because they normally absorb more SWIR light than NIR light (Sun *et al.* 2012). The results show that water surfaces typically indicate low reflectance in the SWIR band and that high reflectance in the green band (Band 3) produces higher MNDWI values.

The MNDWI is an important technique for detecting surface water features and separating them from other land features by applying GIS and remote sensing approaches. The MNDWI values for water bodies are typically greater than 0.02 because the reflectance value interval between Band 3 and Band 11 or the SWIR band is relatively greater than that between the green band (Band 3) and the NIR band (Band 8), which maximizes the optimum threshold value of the MNDWI. The threshold value of 0.02 (Tekeze) to 0.32 MNDWI (Tana) for the selected study areas was introduced as the best possible separation threshold value between land surface features and open water. Compared with non-water surface features or areas, such as vegetation, soil, and built-up features, have higher reflectances in the SWIR 2 (SWIR 2) band and lower reflectances in the green band (Band 3). This results in lower MNDWI values, and similar findings were obtained (Feyisa *et al.* 2014; Gautam *et al.* 2015; Yang & Chen 2017; Wang *et al.* 2018; Laonamsai *et al.* 2023). The accuracy of water surface extraction may be affected by the presence of water hyacinth, limited river width, and variations in water levels between the specified periods (Laonamsai *et al.* 2023). Specifically, Acharya *et al.* (2018) found that NDVI and NDWI showed better results for only pure water pixels with which findings agree.

Unlike the previous indices (NDWI and MNDWI), the WRI has higher values. The majority of WRI values greater than 0.9 for dam reservoirs and natural lakes increase in quality when the water quality increases; however, some WRI values are confused with built-up areas. As an example, the WRI value of Lake Langano approaches the WRI values of built-up areas due to the lower quality of water or being affected by sedimentation. Furthermore, among the mentioned approaches, WRI first and MNDWI were the second most accurate, whereas the least accurate extraction approach was the NDWI. The NDWI and MNDWI were evaluated with various bands of Sentinel-2, and the results suggested that WRI can achieve better water extraction performance; however, three of these methods have difficulty extracting ice-covered surface features (Acharya *et al.* 2018; Khalid *et al.* 2021).

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The findings of this study suggested that for the spatial mapping of surface water bodies for the purpose of extracting pumped hydropower reservoir sites, monitoring water resources through the application of satellite images with surface water extraction indices such as WRI values and the implementation of recommendations to protect and conserve water bodies are highly advised for use as inputs to ecosystems and environmental sustainability (Gautam *et al.* 2015; Khalid *et al.* 2021; Laonamsai *et al.* 2023).

This research advances the field by demonstrating superior performance metrics for WRIs, achieving a PA of 97.98% and a UA of 88.78%, significantly better than indices such as MNDWI and NDWI in the Ethiopian context, especially in areas with limited vegetation cover. This study validates the multiband capabilities of Sentinel-2 for mapping both small and large water bodies with unprecedented accuracy, establishing a reliable methodology for operational water resource monitoring. Furthermore, the research provides detailed accuracy assessments under different environmental conditions, offering valuable insights for selecting appropriate indices on the basis of specific geographical contexts.

Surface water features provide a quick and inexpensive way to map water surfaces by distinguishing them from non-water surfaces. However, seasonal variations may affect performance and produce misleading results. These variations are influenced by turbidity, snow cover, and the presence of seasonal vegetation. For example, during the rainy season, bare land may be covered with vegetation, but not during the dry season. To avoid such confusion, we selected a time range that is free from the above-mentioned factors. In this study, we faced a challenge due to seasonal variations in a small area of Gibe I and III. This seasonal variation caused confusion between water and settlement areas. This can be clearly observed by comparing the OA values of the Gibe I and III NDWI (88.6%), MNDWI (90.2%), and WRI (93.95%) with those of the Central Rift Valley Lakes NDWI (86.78%), MNDWI (94.57%), and WRI (96.37%).

Furthermore, the performance variation between Lake Tana and the Tekeze revealed that the OA values of the Tana NDWI, MNDWI, and WRI were 87.41, 94.61, and 95.13%, respectively, compared with the Tekeze NDWI, MNDWI, and WRI, which were 84.5, 92.7, and 93.73%, respectively. This difference is due to seasonal variations in rainfall and runoff. The rainy season around Lake Tana lasts from May to October, with the highest rainfall occurring from July to September. In contrast, the Tekeze catchment has a shorter rainy season and experiences high sediment accumulation in the Tekeze reservoir during this time.

Recent advances of studies and approaches of geo-information and remote sensing are required in consideration of spatial, temporal, and spectral resolution, and can contribute significantly to extracting surface water bodies for various purposes; however, extraction of surface water bodies still has some challenges due to changes in season, weather changes, and climate changes to implement surface water extraction approaches as was addressed by Tesfaye & Breuer (2024). As for the limitations and challenges of extracting complicated river networks, some areas may be confused with built-up areas, seasonal variation for streams and channels in different seasons, which required more realistic separation of small rivers and streams, might cause some confusion between domestic water and other non-water surface features (Wang *et al.* 2021). The inability to detect certain things is often due to the color of water bodies and the limitations of remote sensing spectral information in complex geographic environments. Mountain shadows are often incorrectly identified as water bodies (Jiang *et al.* 2014; Li *et al.* 2022). Furthermore, there are limitations in discriminating between water and low-albedo pixels, such as shadow and paved surfaces (Feyisa *et al.* 2014).

5. CONCLUSIONS

The study addressed the characterization of optimal thresholds for each site and index to address discrepancies in thresholds affecting surface water detection and achieving well-defined values and approaches. The WRIs were more accurate than MNDWI and NDWI in the detection of surface water. The WRI achieved the highest PA and UA, whereas the MNDWI achieved the second highest PA and UA. Most WRI values greater than 0.9 for surface water features might depict better or clearer water surface features; however, some areas may be confused with built-up areas. WRIs have superior accuracy in surface water detection, and characterized optimal threshold values for each index and study site will help researchers obtain information on water surface features, reservoir and lake levels, changes in water sources, and potential water conflicts. This information is useful for sustainable river management, hydropower and pumped hydropower development, real-time reservoir monitoring, and water resource development and management. Synchronization and integration with the national platform and database will help standardize the national GIS platform by collecting recent ground control points and satellite

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images. This can be used for further training in the manipulation of indices and the use of Sentinel-2 image processing and analysis. Ultimately, integrating WRI into national systems and standardizing site-specific thresholds will lead to more informed decision-making and better stewardship of water resources.

DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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First received 29 December 2024; accepted in revised form 18 August 2025. Available online 30 August 2025

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